

Rotor Model Identification Methodology

S550 hexarotor · maxon ECX 32 flat + APC 10×4.5MR · drone_control_pkgs
Compiled 2026-09-24 · Parameters: motor_const (C_T), moment_const (k_m), rotor command scale, motor response

1. Definition of the rotor model

The controller (control_allocator.py, cmd_converter.py) and the disturbance observers share the following rotor model:

$$T_i = C_T \omega_i^2, \quad \tau_i = k_m T_i = C_T k_m \omega_i^2, \quad \omega_i \text{ [rpm]}$$

where T_i is the thrust of rotor i [N], τ_i its reaction torque [N·m] and ω_i the actual rotor speed [rpm] received over CAN. Rotor commands are sent as the integer cmd_raw (cmd_converter.py, MaxBit = 8191, MaxRpm = 9800):

$$\text{cmd_raw}_i = \left\lfloor \omega_i^{\text{cmd}} \frac{8191}{9800} \right\rfloor, \quad \omega_i^{\text{cmd}} = \text{cmd_raw}_i \frac{9800}{8191}$$

Total force and moments follow from the hexarotor allocation matrix K (arm length $L = 0.265$ m, $l_y = L[1/2, 1, 1/2, -1/2, -1, -1/2]$, $l_x = L[\sin 60^\circ, 0, -\sin 60^\circ, -\sin 60^\circ, 0, \sin 60^\circ]$, $\sigma = [-1, 1, -1, 1, -1, 1]$):

$$f = \sum_i T_i, \quad M_x = \sum_i l_{y,i} T_i, \quad M_y = -\sum_i l_{x,i} T_i, \quad M_z = k_m \sum_i \sigma_i T_i \quad (u = K T)$$

The two constants to identify are C_T and k_m ; the rotor limits (rotor_min/max) and the command-to-speed response are verified from the datasheet and the flight logs.

2. Thrust coefficient C_T

2.1 Method A — steady-hover force balance (primary)

In a level hover the vertical acceleration is zero, so the vertical component of the six rotor thrusts equals the weight. Including the vehicle tilt θ :

$$C_T \sum_{i=1}^6 \omega_i^2 \cos \theta = m g \implies C_T = \frac{m g}{\left\langle \sum_i \omega_i^2 \cos \theta \right\rangle_{\text{hover}}}$$

Procedure

1. Weigh the vehicle independently (3.050 kg after removing the prop guards, 3.322 kg with the payload). A C_T error and a mass error are indistinguishable in this equation, so **the mass must be measured, never fitted**.
2. Select the hover window from the motion-capture altitude, independently of the controller logic: take the interval with $z_{\text{rel}} > 0.05$ m and trim 3 s from each end to exclude the take-off and landing transients.
3. Compute $\Sigma \omega_i^2$ from /uav/actual_rpm (CAN, 100 Hz), multiply by $\cos \theta = R_{33} = 1 - 2(q_x^2 + q_y^2)$ from the odometry quaternion, and average over the window.
4. Evaluate the expression above. Hover tilt is $1\text{--}2^\circ$, so the $\cos \theta$ correction is at the 0.05 % level.

Results

Dataset	m [kg]	mean rpm	C_T (A) [N/rpm ²]	C_T (A, tilt-corr.)	Note
2026-05-15 (prop guards removed)	3.050	≈6200	1.3175e-07	—	value adopted in yaml
DataSet/01 hgdo wo_ff	3.050	6238	1.2816e-07	1.2822e-07	re-identified
DataSet/01 hgdo ff_pivot_based	3.050	≈6240	1.2819e-07	1.2831e-07	re-identified
DataSet/01 l1 wo_ff	3.050	≈6240	1.2847e-07	1.2854e-07	re-identified
DataSet/02 hgdo wo_ff (payload)	3.322	6451	1.3050e-07	1.3059e-07	re-identified

2.2 Method B — HGDO f_z residual matching (cross-check)

The HGDO estimates the residual of the translational dynamics as a disturbance force. Its steady-state mean on the vertical axis is

$$\hat{f}_z \approx m(\ddot{z} + g) - C_T \sum_i \omega_i^2 \cos \theta$$

so, with the measured mass fixed, the C_T that drives the hover mean of $f_{z,DOB}$ to zero must coincide with Method A. On the 2026-05-15 data Method A gave 1.3175e-07 and Method B 1.3183e-07, a **0.06 %** difference. On the present DataSet the explicit-acceleration version (same window as the tilt-corrected column above) gives 1.2825e-07 / 1.3058e-07, within 0.02 % of Method A.

Reverse check (mass recovery): with C_T fixed, the same equation becomes a mass-error estimator. In the payload experiment the HGDO reported $f_{z,DOB} = -3.14$ N in hover, i.e. $\Delta m = 3.14/9.81 = 0.320$ kg, against a measured increase of 0.322 kg (within 0.1 %). This confirms independently that the adopted C_T sets the thrust scale correctly.

2.3 Method C — datasheet comparison

From the bench table of the maxon ECX 32 flat + APC 10×4.5MR combination (ESC 22.2 V, 20 °C, 475 m AMSL), $C_T = T/\omega^2$ and $k_m = \tau/T$ evaluate to:

rpm	Current [A]	Torque [mN·m]	Thrust [g]	C_T [N/rpm ²]	$k_m = \tau/T$ [m]
5900	2.5	68	447	1.260e-07	0.01551
6200	2.9	74	485	1.238e-07	0.01555
6500	3.3	82	539	1.252e-07	0.01551
6800	3.9	91	600	1.273e-07	0.01546
7100	4.5	101	665	1.294e-07	0.01548
7300	5.0	108	716	1.318e-07	0.01538
7600	5.7	117	766	1.301e-07	0.01557
7900	6.6	129	848	1.333e-07	0.01551
8200	7.5	139	920	1.342e-07	0.01540

Table. Continuous-operation rows (winding < 155 °C) of the datasheet and the derived coefficients. Continuous limit 8200–8395 rpm, short-term limit 9900 rpm.

The 7300 rpm row gives $C_T = 1.318$ e-07, essentially the adopted 1.3175e-07. However the datasheet C_T rises monotonically with speed, from 1.24e-07 (5900–6200 rpm) to 1.34e-07 (8200 rpm); at the hover operating point ($\approx$ 6200 rpm) the flight-identified value is 3–6 % above the bench value. The likely causes are (i) residual ground effect at the hover rotor height of 0.315 m ($\approx$ 2.5 R) and (ii) inflow differences between the bench and the installed rotor. **The C_T used for control is therefore identified from flight logs under the same airframe and altitude conditions (Method A);** the datasheet only confirms the order of magnitude and the trend.

2.4 Spread of the values and interpretation

Relative to the adopted value (2026-05-15, 1.3175e-07) the re-identified values are –2.7 % on DataSet/01 and –0.9 % on DataSet/02. A 2–3 % spread is within what day-to-day conditions explain:

- propeller wear and contamination; daily temperature and pressure (air density) changes
- hover-altitude differences through the ground effect ($\approx$ 0.1–0.2 % per cm)
- mass measurement error (± 3 g $\rightarrow$ ± 0.1 %) and the mass drop during a flight (a few grams)
- rotor-speed asymmetry from the CoM offset enters $\Sigma \omega^2$ only at second order

Operating guideline: re-identify C_T with Method A for every campaign (and always after a propeller or frame change); if the hover-mean $f_{z,DOB}$ exceeds ± 0.5 N, suspect C_T or the mass setting first. The yaml currently uses $m = 3.000$ kg (nominal) with $C_T = 1.3175e-07$, so the difference to the true weight (3.05/3.32 kg) is absorbed as a constant disturbance in $f_{z,DOB}$ — intentionally, since it also gives the take-off test $T \geq W_{nom}$ a margin.

3. Reaction-torque coefficient k_m (moment_const = 0.01569 m)

k_m is the ratio of rotor reaction torque to thrust, τ_i/T_i [m], and enters only the yaw moment $M_z = k_m \sum \sigma_i T_i$. The datasheet torque-to-thrust ratio is nearly constant, 0.01538–0.01555 m over 5900–8200 rpm, and the adopted 0.01569 m is within about 1 % of its mean (0.0155 m).

Identifying k_m from flight logs alone would require the yaw dynamics $J_{zz} d\omega_z/dt = k_m \sum \sigma_i T_i - \tau_{drag}$, but the hover yaw moment is only ≈ 0.006 N·m (std), so the signal-to-noise ratio is poor. It is therefore practical to **adopt the datasheet ratio and only verify sign and magnitude from the yaw response** (no divergence or reversal). If a precise value is needed, apply a yaw ramp moment on the ground and measure the reaction torque directly with the load cells used for the moment_offset identification.

Record note: the procedure by which 0.01569 was originally set is not documented in the retained logs or scripts; the datasheet comparison above is the a-posteriori justification of the value.

4. Motor/ESC command response (cmd_raw → actual_rpm)

The rotor-speed model used in simulation and in the NMPC prediction is a second-order lag with a pure delay,

$$H_{rot}(s) = \frac{\omega_{rot}(s)}{\omega_{cmd}(s)} = \frac{K \omega_n^2 e^{-\tau_d s}}{s^2 + 2\zeta \omega_n s + \omega_n^2}, \quad K = 1 \text{ (nominal)}$$

whose parameters (ω_n , ζ , τ_d) are identified from the commanded speed $x = \text{cmd_raw} \cdot 9800/8191$ and the measured speed $y = \text{actual_rpm}$. This section derives the estimator, states the practical settings, and discusses why different identification runs land on different but nearly equivalent parameter triples.

4.1 Derivation of the cross-spectral estimator $H = S_{xy}/S_{xx}$

Step 1 — LTI system with output noise. With mean-removed signals, the motor/ESC is modelled as a linear time-invariant system with impulse response $h(t)$, plus measurement noise $n(t)$ that is uncorrelated with the input:

$$y(t) = \int_0^\infty h(u) x(t-u) du + n(t), \quad E[x(t) n(t+\tau)] = 0$$

Step 2 — correlation functions. Multiplying by $x(t-\tau)$ and taking expectations (stationarity assumed), the noise term vanishes and the cross-correlation becomes the auto-correlation convolved with h :

$$R_{xy}(\tau) := E[x(t) y(t+\tau)] = \int_0^\infty h(u) R_{xx}(\tau-u) du = (h * R_{xx})(\tau)$$

Step 3 — frequency domain (Wiener–Khinchin). Since the Fourier transform of a correlation function is the corresponding spectral density, the convolution becomes a product:

$$S_{xy}(f) = H(f) S_{xx}(f) \implies H(f) = \frac{S_{xy}(f)}{S_{xx}(f)}$$

This is the H_1 estimator. It is preferred over the direct ratio $Y(f)/X(f) = H + N/X$ because only the part of y that is correlated with x survives in S_{xy} ; the noise averages out instead of being carried through.

Step 4 — finite-record estimate (Welch). The record is split into K overlapping segments of length L (Hann window w , 50 % overlap); with segment spectra $X_k(f)$, $Y_k(f)$:

$$\hat{S}_{xx}(f) = \frac{1}{K} \sum_{k=1}^K \frac{|X_k(f)|^2}{f_s \sum w^2}, \quad \hat{S}_{xy}(f) = \frac{1}{K} \sum_{k=1}^K \frac{X_k^*(f) Y_k(f)}{f_s \sum w^2}, \quad \hat{H}(f) = \frac{\hat{S}_{xy}}{\hat{S}_{xx}}$$

The normalisation cancels in the ratio. The validity of each frequency bin is judged by the coherence

$$\gamma^2(f) = \frac{|\hat{S}_{xy}(f)|^2}{\hat{S}_{xx}(f)\hat{S}_{yy}(f)} \in [0, 1],$$

which is close to 1 where y is linearly explained by x and drops where noise, non-linearity or other inputs dominate; bins with $\gamma^2 < 0.5$ are excluded from the fit and the others are weighted by γ^2 .

Settings used. $f_s = 100$ Hz (CAN rpm), $L = 512$ (0.195 Hz resolution), hover window from the motion-capture altitude with 3 s trimmed at each end, six rotors pooled by summing the estimates S_{xx} , S_{xy} , S_{yy} before dividing (lower variance than averaging six separate ratios). Coherence is 0.99 (median) over 0.2–5 Hz and stays above 0.5 up to ≈ 12 Hz.

4.2 From the estimated $H(f)$ to $(K, \omega_n, \zeta, \tau_d)$

Substituting $s = j\omega$ in the model separates magnitude and phase:

$$|H| = \frac{K \omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}}, \quad \angle H = -\omega\tau_d - \text{atan2}(2\zeta\omega_n\omega, \omega_n^2 - \omega^2)$$

The delay does not affect the magnitude, so (K, ω_n, ζ) are determined by the magnitude $|H|$ and τ_d by the phase remaining after the second-order phase is removed. In practice all four are fitted jointly by weighted least squares on the complex residual (which enforces magnitude and unwrapped phase together):

$$\min_{K, \omega_n, \zeta, \tau_d} \sum_{f_k: \gamma^2 > 0.5} \gamma^2(f_k) |H_{\text{model}}(f_k) - \hat{H}(f_k)|^2$$

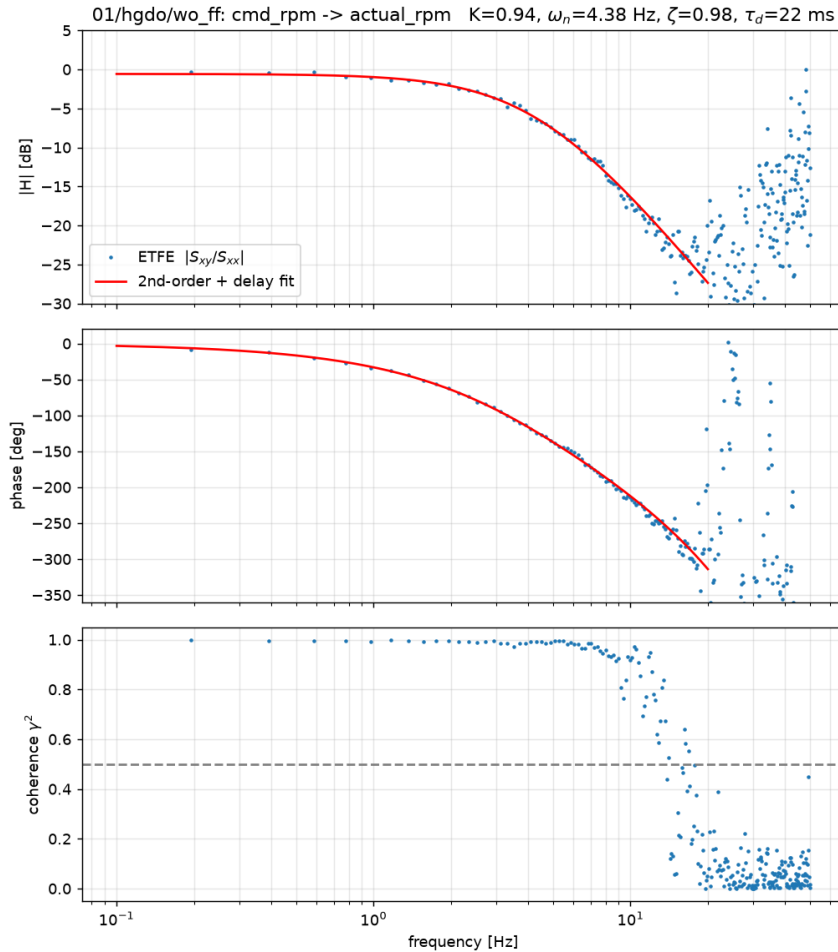


Figure. Pooled ETFE (dots) and second-order-plus-delay fit (red) for DataSet/01 hgdo wo_ff; bottom: coherence.

Dataset	K	ω_n [rad/s]	f_n [Hz]	ζ	τ_d [ms]	f(-3 dB) [Hz]
DataSet/01 hgdo wo_ff	0.936	27.5	4.38	0.98	22	2.93
DataSet/02 hgdo wo_ff	0.942	27.6	4.39	1.01	23	2.73
with K fixed to 1 (01, 0.2–12 Hz)	1	27.6	4.39	1.05	22	—

Table. Fitted parameters. The two flights agree closely; $K < 1$ is attributed to the closed-loop bias discussed in 4.4 and $K = 1$ is used in the model.

Interpretation: $\zeta \approx 1$ (critical damping) means the two poles nearly coincide at $-\omega_n$, i.e. two first-order lags of ≈ 36 ms in series; the -3 dB frequency of a critically damped second-order system is $0.64 \omega_n = 2.8$ Hz, matching the directly measured 2.7–2.9 Hz. The pure delay of ≈ 22 ms is the CAN transport plus ESC processing.

4.3 Why an earlier identification gave $(\zeta, \omega_n, \tau_d) = (0.78, 21.7 \text{ rad/s}, 15.7 \text{ ms})$

The three parameters are strongly correlated: along a ridge in parameter space, a larger ζ slows the corner, a larger ω_n compensates it, and a larger τ_d restores the phase lost by the second-order term, so the frequency response is almost preserved. Fixing ζ and re-fitting the other two on the present data traces this ridge:

ζ (fixed)	best ω_n [rad/s]	best τ_d [ms]	weighted residual
0.70	20.4	11.2	7.7e-3
0.78	21.9	14.2	4.1e-3
0.90	24.3	18.1	1.6e-3
1.00	26.5	20.7	9.6e-4
1.10	28.8	23.0	9.5e-4

Table. Identifiability ridge on DataSet/01 ($K = 1$, 0.2–12 Hz). The earlier triple lies on the $\zeta = 0.78$ slice of the same ridge (best point 21.9 rad/s, 14.2 ms vs. reported 21.74 rad/s, 15.67 ms).

The two models differ by less than 0.5 dB and 7° below 4 Hz and only diverge in the roll-off above 6 Hz (the earlier set rolls off faster but has a shorter delay). For attitude control (1.6 Hz band) and for the cmd_lpf setting (7 Hz) they are interchangeable. Which point of the ridge an identification lands on depends on:

- **fitting band** — with $K = 1$ the fit over 0.2–3 Hz alone gives (49 rad/s, $\zeta = 1.7$, 30 ms), over 0.2–5 Hz (29 rad/s, 1.1, 22 ms), over 0.2–12 Hz (28 rad/s, 1.05, 22 ms): without content above ≈ 5 Hz the triple is not identifiable and the excitation bandwidth decides the result;
- **fitting method** — a time-domain output-error fit weights the low-frequency energy, a frequency-domain complex-residual fit weights bins evenly within the band;
- **treatment of K** — fixing $K = 1$ shifts the remaining three slightly;
- **plant changes** between campaigns (prop guards, battery voltage, ESC settings) are possible but not required to explain the difference — on the present data the earlier triple has a 10× larger residual than the new fit, so the new values describe the current airframe better.

Recommendation: report ridge-insensitive invariants alongside the triple — the -3 dB frequency (3.1 Hz for the earlier set, 2.9 Hz here) and the phase at 4 Hz (-123° vs -116°) — and, for a definitive value, re-identify open-loop with a 0.5–20 Hz chirp on cmd_raw with the vehicle fixed on the ground, stating the fitting band and the K convention.

4.4 Closed-loop caveat

In hover logs x is the controller output and the controller feeds back states derived from y , so the assumption $E[x \, n] = 0$ does not hold exactly. The H_1 estimator then tends towards

$$\hat{H} \rightarrow \frac{HS_{rr} - C^{-1}S_{nn}}{S_{rr} + |C|^{-2}S_{nn}},$$

with r the exogenous excitation and C the equivalent controller: it is biased towards $-1/C$ where noise dominates. With coherence 0.99 the excitation dominates here and the bias is small (it is the likely origin of $K = 0.94$), but an open-loop measurement is the reference method for a publication-grade value.

5. Rotor limits and derived constants

Item	Value	Source / note
rotor_min	2000 rpm $\rightarrow$ $T_{\min} = 0.527$ N	yaml; idle stability
rotor_max	7900 rpm $\rightarrow$ $T_{\max} = 8.22$ N	yaml (raised 7300 $\rightarrow$ 7900); cont. limit 8200 rpm
hover operating point	$\approx$ 6200–6450 rpm, 5.0–5.4 N per rotor	weight 3.05–3.32 kg
upward margin (7900)	3.21 N per rotor	$\Delta_{\max}$ in the moment-authority analysis
motor_const C_T	$1.3175e-07$ N/rpm ²	Method A, 2026-05-15
moment_const k_m	0.01569 m	within 1 % of datasheet τ/T
command scale	8191 / 9800	cmd_converter.py

6. Reproduction checklist

1. Weigh the vehicle $\rightarrow$ 2. record ≥ 20 s of hover (/uav/actual_rpm, /uav/cmd_raw, /mavros/local_position/odom) $\rightarrow$ 3. run `python3 DataSet/_rotor_model_id.py <bag> <mass>` $\rightarrow$ 4. write the (A') value into motor_const and synchronise the three yaml files (pd_nmpc_att_with_hgdo/l1, hgdo_config, l1_adaptive_config) $\rightarrow$ 5. on the next flight, verify that the hover-mean $f_{z,DOB}$ equals $(m_{\text{true}} - m_{\text{nom}}) \cdot g$.

The script selects the hover window from the motion-capture altitude (independent of the controller decision), prints the tilt-corrected and acceleration-inclusive variants, and computes the motor ETFE (DC gain, -3 dB, equivalent delay) on the same window.

Related files: drone_control/config/*.yaml, drone_dob/config/*.yaml, drone_control/drone_control/utils/{cmd_converter, control_allocator}.py, DataSet/_rotor_model_id.py